

PROJECT ASTRA: AUTONOMOUS TACTICAL RECONNAISSANCE SYSTEM (ATARS)
DEFENCE DOCUMENT INTELLIGENCE SAMPLE SPECIFICATION - UNCLASSIFIED
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1.0 EXECUTIVE OVERVIEW & MISSION SCOPE

Project Astra is an advanced unmanned aerial reconnaissance platform engineered to deliver persistent intelligence, surveillance, target acquisition, and reconnaissance (ISTAR) capabilities across contested multi-domain theatres.

The system addresses the operational requirement for deep-penetration tactical awareness without exposing human crew to integrated air defence systems (IADS).

1.1 SYSTEM CONCEPT & OPERATIONAL ARCHITECTURE

The ATARS airframe utilizes carbon-fiber composite materials offering a low radar cross-section (RCS). Equipped with an autonomous edge-compute processing node, the platform performs real-time automated target recognition (ATR) directly on-board, minimizing transmission bandwidth back to ground control.

2.0 PROPULSION & ENDURANCE

- Powerplant: Hybrid hydrogen-electric fuel cell with regenerative turbo-alternator.
- Operational Flight Endurance: 36 hours continuous loiter time.
- Maximum Service Ceiling: 45,000 feet above mean sea level (MSL).
- Cruising Speed: 180 knots (approx 333 km/h).
- Maximum Dash Speed: 260 knots.

2.1 SECURE COMMUNICATIONS & DATA LINKS

The system maintains communications through redundant line-of-sight (LOS) and satellite communications (SATCOM):

- Ku-Band and Ka-band secure satellite uplinks with automated anti-jamming beam steering.
- Encryption: Hardware-level AES-256-GCM authenticated telemetry.
- Frequency-hopping spread spectrum (FHSS) operating between 12 GHz and 18 GHz.

3.0 MAJOR APPLICATIONS

The Astra system is configured to fulfill four primary operational defence roles:

1. Maritime Border Surveillance: Continuous monitoring of littoral waters and identification of non-flagged surface vessels.
2. Artillery Fire Direction: Rapid forward observer targeting and battle damage assessment (BDA).
3. Tactical Electronic Warfare (EW): Signals intelligence (SIGINT) emitter detection and localized GPS denial protection.
4. Search and Rescue (SAR): High-resolution infrared scanning during severe weather operations.

3.1 SENSOR ARCHITECTURE

- High-Definition Electro-Optical/Infrared (EO/IR) gyrostabilized turret with 50x optical magnification.
- Multi-mode Synthetic Aperture Radar (SAR) capable of ground moving target indication (GMTI).
- Long-range LiDAR scanner for sub-meter 3D terrain topography mapping.

4.0 ENVIRONMENTAL & WEATHER LIMITATIONS

- Operating Ambient Temperature Range: -40 degrees Celsius to +55 degrees Celsius.
- Maximum Crosswind Landing Limit: 35 knots.
- De-icing System: Electro-thermal leading edge wing heating.

4.1 LOGISTICS & OPERATIONAL TURNAROUND

- Base Turnaround Cycle: 90 minutes between recovery and redeployment.
- Mean Time Between Failures (MTBF): Certified at 1,200 operational flight hours.
- Crew Requirement: 2 operators per ground control station (one mission commander and one payload specialist).